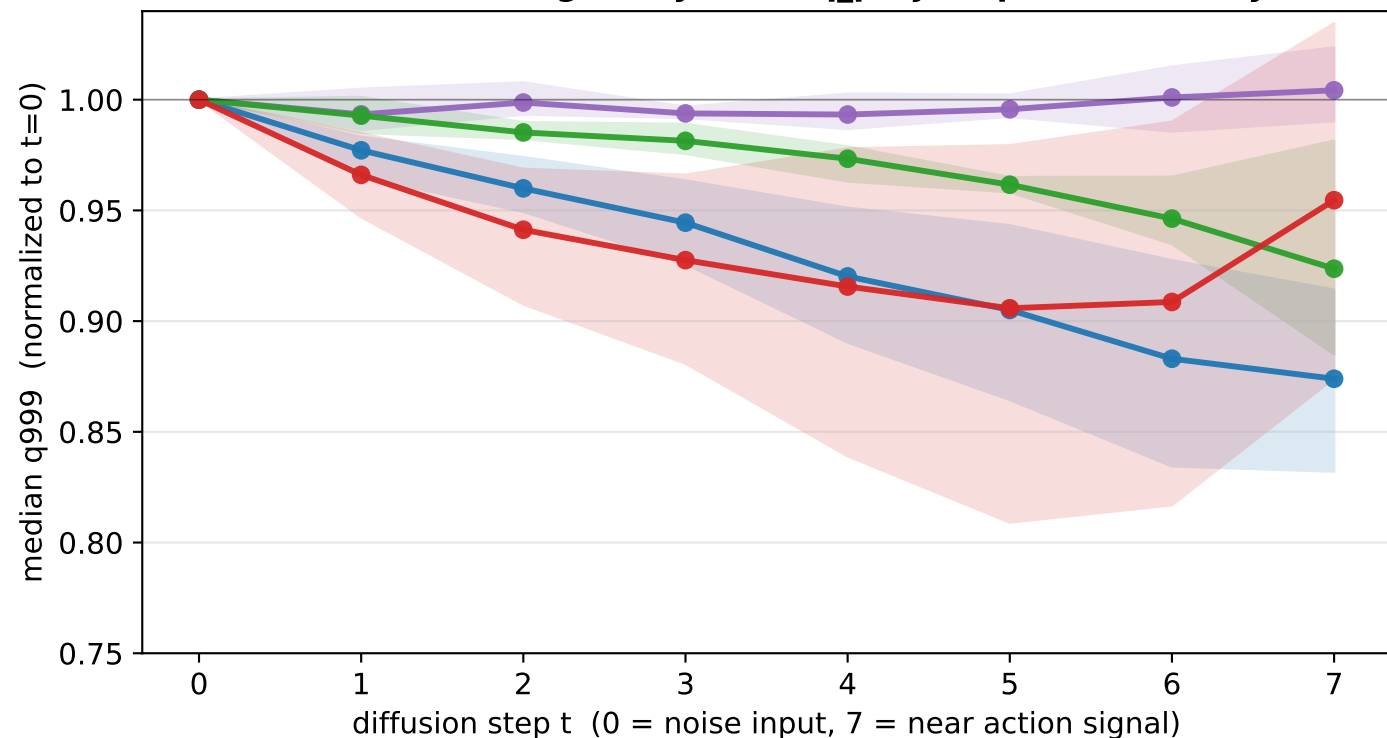


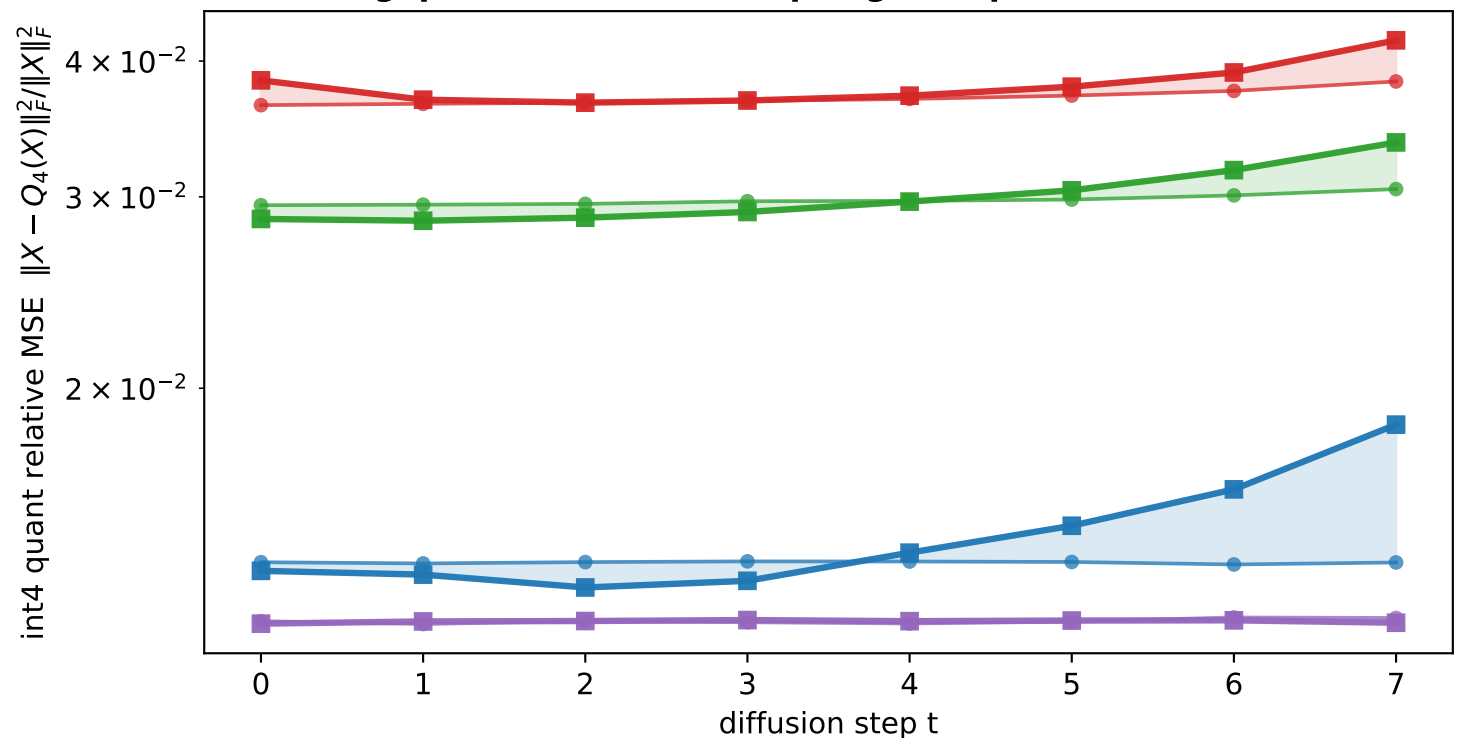
Per-step act_scale necessity — empirical motivation (gr00t-N1.5 DiT, libero_object cal)

Panel 1 — q999 drift across 8 diffusion steps
Monotone decreasing; early-block q_proj drops ~17-20% by t=7



● q_proj (attn input, q/k/v shared X)
● o_proj (attn output, post-softmax)
● ff_net_0 (MLP input)
● down_proj (MLP output, post-GLU)

Panel 2 — Per-step vs single-bucket int4 quant MSE
Shaded gap = waste from collapsing 8 step-scales into one bucket



● q_proj (attn input, q/k/v shared X) per-step scale
● o_proj (attn output, post-softmax) per-step scale
■ q_proj (attn input, q/k/v shared X) single-bucket
■ o_proj (attn output, post-softmax) single-bucket
● ff_net_0 (MLP input) per-step scale
● down_proj (MLP output, post-GLU) per-step scale
■ ff_net_0 (MLP input) single-bucket
■ down_proj (MLP output, post-GLU) single-bucket